EPSR-test-ex51 partials: centre atom, rho(z) and rho(r)

Program(s): partials | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the three restricted modes of `partials` [1]: the g(r) of all atom types around one centre type (`atom-c`), the density profile along z about a centre type (`atom-z`) and the density versus distance in the xy-plane about a centre type (`atom-r`), together with the no-accumulation branch (`nsumt -1`) and the range scaling `rnormalise`.

Code exercised

Fortran (original, fortran_src/):

- `partials.f`: `run_partials`, `find_partials` (centre-atom, rho(z) and rho(r) branches)
- `read_and_save_files.f`: `read_input_file`, `initgofz`, `outgofz` (histogram file, running average and spread), `save_files_others`; `auxiliary_routines.f`: `r_int`
- `setup_symbols_generic.f`: `setup_symbols_partials`

Rust (port, rust/epsr-core/src/):

- `tools/partials.rs`: `run`, `find_partials`
- `tools/anal_common.rs`: `match_centre`, `outgofz`, `write_fort14`
- `symbols.rs`: `setup_input_file`, `read_input_file`, `save_inp_file`; `output.rs`: `r_int`

Physical system

Analysed configuration: `200DMSO400Water.ato`, an unchanged copy of the EPSR-test box of 200 dimethyl sulphoxide (DMSO; atom types S, O, C, H) and 400 water molecules (O1, H1): 3200 atoms, DMSO mole fraction 1/3, 300 K, cubic box of side 33.1213 Å (0.0881 atoms/Å^3), the EPSR model of the deuterated DMSO-water neutron data (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), cf. [2]. `centreO1`: the 6 g(r) of every type around water O1, normalised with the O1 density, 200 points out to 0.8 of the default range; `rhozS`: number density versus the z separation from DMSO S atoms, 150 points; `rhorO`: number density versus the distance in the xy-plane from DMSO O atoms, 100 points over half the default range (`rnormalise 0.5`). Such profiles characterise the local environment of one site [3].

Inputs

Three keyword files: `centreO1.PARTIALS.dat` (`atom-c O1`, `nz 200`, `rnormalise 0.8`, `nsumt -1`), `rhozS.PARTIALS.dat` (`atom-z S`, `nz 150`, `nsumt 0`), `rhorO.PARTIALS.dat` (`atom-r O`, `nz 100`, `rnormalise 0.5`, `nsumt 0`); `200DMSO400Water.ato` copied unchanged from EPSR-test.

How the accuracy was verified

`commands.txt` (3 program calls, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
`commands.txt` runs the three inputs in sequence and the whole list twice. `centreO1` (`nsumt -1`) writes the single-configuration result in both passes and its `nsumt` stays -1; `rhozS` and `rhorO` accumulate (reference after pass 2: `nsumt 2`). Compared: the three `.PARTIALS.g01` files, the three rewritten `.dat` files and `fort.14`, which every run overwrites, so the one of `rhorO` is kept: it lists only the 6 pairs O-S ... O-H1 with O as centre.

Notes

`fort.14`: the Fortran writes these lines to unit 14 after `save_files_others` has closed it, so gfortran creates the default file `fort.14`; the port writes the same file.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[3] J.-P. Hansen and I. R. McDonald, Theory of Simple Liquids, 3rd ed., Academic Press (2006).
[4] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.